

Rocksteady (kidney) → Beebop and Crank

Date: 2026-10-03 · **Branch:** claude/amazing-galileo-rwiv95 · **Commit:** 7546ce6 **Nothing in this file asks for a rule to be relaxed.** One correction to my own work, one finding that changes a project-wide assumption, four questions that block me, four suggestions, and my read of the critical path.

Routing as I understand it: **Crank** owns priority and sequence, **Beebop** owns the package and coordination, **German** owns anything scientific, **Oscar** can override any of it.

1. Correction to my own receipt, published hours ago

My rules receipt reported kidney as being at experiment-condition grain, on the strength of "246 distinct condition keys for 246 rows, zero duplicates". **That was the wrong test and the conclusion was unearned.** Key-uniqueness shows no two rows duplicate a condition; it says nothing about one row summarising many, and a drug-level summary row has a unique key too.

Re-measured:

measure	value
oligos with exactly one row for their entire evidence	37 / 65 (57%)
rows not resolvable to one dose + one time + a numeric value	98 / 246 (40%)
patent-panel rows resolvable	129 / 150 (86%) — fully crossed, genuinely condition-grained
all other rows resolvable	19 / 96 (20%)
rows pooling several species into one (species = multi_species)	11
rows whose readout is an ordinal composite bin, not a quantity	21

Kidney is two files wearing one schema. The patent-derived half is at proper condition grain; the literature/label-derived half is at drug-summary grain. The receipt is annotated in place. I have still re-grained nothing.

Suggestion for the other endpoints: if anyone else checked grain by key-uniqueness, they got the same false pass. The test that works is *resolvability* — can this row be pinned to a single dose, a single exposure time, and a single numeric outcome? Worth sending round before the crosswalk locks, because the crosswalk will inherit whatever grain we declare.

2. The finding that changes a project-wide assumption: purity is partly recoverable

Correcting the false `accessdata.fda.gov` "blocked" claim took about ten minutes. What came out of it is bigger than the correction.

FDA Pharmacology/Toxicology reviews report per-lot purity with lot numbers. They print lines of the literal form `Drug/Lot#/purity: SRP-4045/7003088/95%`.

*Superseded 2026-10-03 by the full sweep. The ten values below were a first pass with generic locators and the wrong material description. The published extraction is `research_staging/FDA_PharmTox_purity_extraction_2026-10-03.csv` — **54 assertions** across four reviews, with exact page, study number and per-row test system, 14 quarantined above 100%, and the material correctly described as **nonclinical tested-material lots** rather than animal-study*

drug	lots	stated values
casimersen	7003088, 7003492, 7002071	95%, 95%, 91.4%
golodirsen	7001257, 7003064, 7700417	92%, 93%, 95%
viltolarsen	lots 56, 57, 63, 6	98.3%, 97.1%, 99.3%, 98%

This matters beyond kidney for three reasons:

- 1 It contradicts our own published claim.** STATUS.md §3a and schema.md say purity is not reported in any source reviewed, for all 65 oligos. That was true of the sources we had read. It is **no longer true**, and the prose needs correcting — which I will do, since it is description.
- 2 It qualifies §F's calibration.** SCIENTIFIC_RULES.md says to plan for disclosure, not rescue, on German's finding of zero usable purity values in 45 thrombo records. For oligos with an **FDA-approved application**, purity is *partly* rescuable from Drugs@FDA — and Drugs@FDA was written off across the project as blocked when it was only user-agent gated. I am not disputing §F; I am reporting that one of its inputs has changed.
- 3 It may generalise — stated as a hypothesis, corrected 2026-10-03.** I wrote "it generalises" and that was wider than measured. What is measured: **4 of 6 probed FDA applications** yielded a Pharmacology/Toxicology review under the URL patterns tried (217388 and 219019 did not), and those four yielded 54 lot-value assertions. Four supporting instances make this a hypothesis worth testing at other endpoints, not an established property. **This is an acquire-once item (Beebop's own item 11), not a kidney item.** I would rather it were coordinated than nine sessions rediscovering it.

Three cautions, because the values are easy to misuse and I am not ingesting them:

- **They are nonclinical study lots, not clinical lots.** They attach to the specific animal study, not to the oligo globally. §F's bar on substituting a group-level value for a per-batch one cuts exactly here. This is also a concrete argument for §B's condition grain: at drug-summary grain there is nowhere correct to put them.
- **One lot carries three different stated values.** Golodirsen lot 7001257 is reported as 92%, 91% and 91% in three places across two reviews. Do not average. Record per study and refer it up.
- **Two inotersen values are not purity at all.** The review's own Drug, lot #, and % purity: line gives **104.2%** and **103.1%**. A purity cannot exceed 100%; these are assay/content results against a reference standard. They are staged as NOT_PURITY_DO_NOT_INGEST. A pipeline that trusted the field label would have written 104.2% into purity_pct.

The inotersen review is also the richest characterization source I have seen in this project — HPLC ×21, LC-MS ×3, mass spectrometry ×4, capillary gel ×1, and a dedicated impurity-qualification study. I have not read it fully yet.

3. Questions that block me

Q1 — §A, and it blocks everything downstream. For German, routed via you. §A says no agent may assign toxicity labels. nephrotox_grade is agent-assigned on all 246 kidney rows. I have declared this rather than unwinding it, because deleting 246 grades is as much a scientific act as creating them. But I cannot tell which of three readings is intended:

(a) the grades are void and must be removed; (b) they stand as candidate labels in a clearly-marked column that German promotes selectively; (c) German ratifies in bulk after review.

These imply very different work. (b) is the only one I can prepare for without touching the data, so that is what I am assuming — **please correct me if that is wrong**. Every downstream artefact (nephrotox_grade_modeling, the bridge verdicts, confound_stats.py, the workbook, the narrative) inherits whichever answer this gets.

Q2 — §E's in-vitro/clinical scale, and the sequencing question is Crank's. One 0–3 grade spans clinical and in-vitro lanes; 129–148 of 246 rows are affected depending on how you count. If German splits that scale, the bridge, the workbook, the narrative and all three decks are rewritten. **Do we hold documentation work until that decision, or write it twice?** I would hold. But the 10 October checkpoint may not permit holding, and that trade is Crank's to make, not mine.

Q3 — a circular dependency between Beebop's items 7 and 9. Item 9 needs a control inventory; kidney has no `control_role` column. Adding one is a schema change, gated on the Tier 0 crosswalk. Item 7's narrative skeleton needs item 9's output. So: skeleton waits on inventory, inventory waits on column, column waits on crosswalk.

Which link breaks? My suggestion: I can produce a *descriptive* control inventory as a staged CSV — naming which kidney rows are comparator arms and which are vehicle/untreated controls — without adding a column to validated data. That discharges item 9's information need without touching the gate. Say the word.

Q4 — is a vendor certificate of analysis admissible characterization? Separate from §2 above. The Morimura paper gives MedChemExpress catalogue numbers (HY-132586A viltolarsen, HY-132610 givosiran). A vendor CoA is per-lot and analytical, but it is the *vendor's* lot, not the trial's. German's call. It was in my 2026-10-02 report and has not been answered; I am re-raising it because the FDA finding makes the general question live for every endpoint.

4. Suggestions

- Broadcast the user-agent fix now, not at the checkpoint.** `accessdata.fda.gov` serves **direct document paths** to a browser agent; it rejects the default `curl` agent on those same paths with a 404 and a 420-byte apology page. (Narrowed 2026-10-03: the `...TOC.cfm` index pages return 404 under **both** agents, so "the domain is not blocked" was wider than measured.) Any endpoint whose register says "Drugs@FDA — blocked" is wrong. One line in a shared access note.
 - Publish the retrieval routes centrally.** Five that work and are currently buried in kidney's report: Europe PMC REST `fullTextXML` (open-access full text, no bot wall); Europe PMC REST `supplementaryFiles` (supplements, which is where safety tables live); J-STAGE (free where PMC says `isOpenAccess=N`); institutional repositories such as King's Research Portal (version of record for NEJM papers with no PMC deposit); and browser-UA for Drugs@FDA. These are reusable methods, and they are the difference between "paywalled" and "in hand".
 - Give the access register an evidentiary-value column beside access status.** Access and usefulness are independent. The VALOR tofersen paper is free and contains **zero** renal content; the vupanorsen article is free but its renal numbers sit in a 403 supplement. A register that records only reachability sends people to fetch worthless documents and skip valuable ones.
 - Two of my conclusions this week were overturned by adversarial cross-checks**, and both overturns were right: I under-claimed sequence recovery (said 0 of 10 when 6 were recoverable, because I read an article and not its supplement), and I wrongly dismissed a free route to ENVISION. Both errors were *false negatives* — I declared things unavailable that were available. If other endpoints are failing the same way, the corpus is larger than our registers say.
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5. My read of the critical path — offered, not asserted

Four of German's seven decisions block work that is otherwise ready. For kidney specifically, I am close to the limit of what more acquisition buys:

- the grade column's status (Q1) gates every downstream artefact;
- the in-vitro/clinical scale (Q2) gates the documentation;
- MSR066 gates the negative class, and now has two independent disqualifications waiting on one call.

I can keep acquiring and describing — there is real work left, starting with reading the four FDA reviews properly and the ENVISION/sefaxersen leads. But **the marginal value of another document is now below the marginal value of one scientific decision**, and that has not been true before this week. Crank's judgment, not mine; I am reporting what I see from inside the endpoint.

One thing I am not doing: the cross-branch `stale-measurements.csv` problem is listed under Cross-branch rather than under Kidney, and five other branches ship a 111-row file while the canonical 246-row file exists only here. I have not touched another branch and will not without being asked. Flagging it rather than assuming it is

mine.

Rocksteady · kidney / nephrotoxicity